Irredundant generation and finite cores on trees

Human Compression project

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Abstract

We combine bounds on maximum irredundant generating length with an exact attachment theorem for permutation groups. Normal-module and simple-factor arguments give a uniform bound for every transitive imprimitive group of degree at least nine. Primitive groups require an explicit classical classification input and a bounded exact catalogue calculation. A separate finite-core interface proves all subgroup intersections after attaching an arbitrary tree. Finally we give a point-stabilizer proof of the tree presentation of a symmetric group and a general change-of-marking rule. The companion application derives the complete high-rank symmetric-group classification, its geometries and its presentations, with an explicit correction to a printed branching clause. All retained hypotheses and the limited formal scope are recorded here.

A marking is a finite indexed generating family of distinct nonidentity involutions. It is a *C-group marking* if, for every pair of subsets S, R of its index set,

$$\langle \rho_i : i \in S \rangle \cap \langle \rho_i : i \in R \rangle = \langle \rho_i : i \in S \cap R \rangle.$$

In particular every subfamily is independent: a generator cannot lie in the subgroup generated by the others. Its rank is the number of indices. The results on irredundant generation below apply to arbitrary finite groups, not only to involution markings. All trees and underlying permutation sets in the attachment and presentation sections are finite.

1 Maximum irredundant length and a normal module

A generating family is irredundant if deleting any member leaves a proper subgroup. Write $\mu(G)$ for the largest size of an irredundant generating family of a finite group G, and $\mu'(G)$ for the largest size of any independent family in it. These are maximum lengths, not the minimum number of generators. For an abelian normal subgroup $V \trianglelefteq G$, let $\mu'_G(V)$ denote the largest size of a family independent under generation by all G-conjugates.

Lemma 1. *For finite G and $N \trianglelefteq G$, $\mu(G) \le \mu(G/N) + \mu'(N)$. If N is abelian, the final term can be replaced by $\mu'_G(N)$. In particular, if $V \trianglelefteq G$ is an elementary abelian minimal normal subgroup, then*

$$\mu(G) \le \mu(G/V) + 1. \tag{1}$$

No splitting of the extension is needed.

Proof. Take an irredundant generating family S of G, and an inclusion-minimal subfamily T whose images generate G/N. For $s \in S \setminus T$, choose a word w_s in T with $sN = w_s N$, and put $v_s = s w_s^{-1} \in N$. If one v_s lies in the subgroup generated by the others, substitution expresses s using $S \setminus \{s\}$, a contradiction. Since $|T| \le \mu(G/N)$, the first bound follows.

When N is abelian, $G = N\langle T \rangle$ implies that its conjugation action on N is already the action of $\langle T \rangle$. Thus an expression for v_s in conjugates of the other v_t again substitutes to an expression for s without s. This proves the normal-module bound. If V is elementary abelian minimal normal, it is an irreducible $\mathbb{F}_p G$-module. Any nonzero vector generates all of V under conjugation, so an independent family for this action has at most one member. This proves (1) for maximum irredundant length. □

This is Cameron–Cara's normal-quotient theorem [CC, Theorem 1.1], previously included as Proposition B2.RelativeBound in our forest theory. We repeat the entire argument, including its action-independent form, so the present common paper has no unpaid dependence on that edition. We retain Whiston's bound that an independent family in S_m has at most $m-1$ members, with equality only when it generates S_m. In particular $\mu'(A_m) \le m-2$. No classification of the C-groups of PSL_2 or PGL_2 is used below.

Corollary 2. *If G has a faithful primitive affine action with elementary abelian translation subgroup V, then* (1) *applies. In particular $\mu(\mathrm{AGL}(4,2)) \le 7$.*

Proof. The translation module is irreducible: a proper invariant subspace would give a nontrivial block system. Thus V is minimal normal. For $\mathrm{AGL}(4,2)$ the quotient is $\mathrm{GL}(4,2) \cong A_8$, so Whiston gives $\mu(A_8) \le 6$. The classical exceptional isomorphism is a retained identification; the finite certificate also checks an explicit isomorphism for the catalogue representation used below. □

2 All transitive imprimitive groups of degree nine

Theorem 3. *Every transitive subgroup of $S_3 \wr S_3$ in its imprimitive action on nine letters has $\mu(G) \le 5$. Consequently it cannot be a maximal parabolic of a rank-seven C-group marking of S_9.*

Proof. Let K be the kernel of the action on the three blocks, and put $A = K \cap A_3^3$. Then A is an elementary abelian normal 3-subgroup, viewed as an invariant subspace of $\mathbb{F}_3^3$. The quotient G/A has normal subgroup $W = K/A \le C_2^3$ and transitive quotient $B \le S_3$, so $B = C_3$ or S_3.

Choose an element of G inducing a three-cycle on the blocks. On A_3^3 its conjugation action is a cyclic permutation of the coordinates with coefficients $\varepsilon_i \in \{1,-1\}$. The vector in any one coordinate is cyclic for this operator, whose characteristic and minimal polynomials are $X^3 - \delta = (X-\delta)^3$, where $\delta = \varepsilon_1\varepsilon_2\varepsilon_3 \in \mathbb{F}_3$. Thus the full module is $\mathbb{F}_3[X]/(X-\delta)^3$. Its invariant subspaces form a chain, and each is cyclic. Any family of vectors in an invariant subspace therefore has one vector whose cyclic span contains all the others. Independence under the whole group implies independence under this one element, so $\mu'_G(A) \le 1$.

The corresponding action on C_2^3 is ordinary coordinate permutation. Under the three-cycle it is the direct sum of a trivial line and the irreducible plane with polynomial $X^2 + X + 1$. Every invariant subspace is a sum of a subset of these two nonisomorphic irreducibles. The two polynomial projection operators show that a vector generates each component on which it is nonzero. At most two vectors are therefore needed from any given family to generate its cyclic submodule; a family independent under the full group has at most two members. Hence $\mu'_{G/A}(W) \le 2$.

Twice applying Lemma 1 gives

$$\mu(G) \le \mu(G/A) + 1 \le \mu(B) + 2 + 1 \le 5.$$

This applies to the maximum irredundant length, not just a selected small generating set. A parabolic of rank six in a C-group has six independent distinguished generators, contradicting the bound. □

3 Independent vectors in a transitive binary module

The following elementary observation is useful beyond the present classification. It bounds irredundance of a particular family, rather than merely providing some short generating family for the module.

Lemma 4. *Let M be a finite-dimensional module over a finite field, generated irredundantly as a module by $v_1, \dots, v_s$. There are simple quotients $L_1, \dots, L_s$ of M and a surjection $M \longrightarrow \bigoplus_{i=1}^{s} L_i$.*

Proof. For each i, choose a maximal proper submodule M_i containing all v_j with $j \neq i$. It does not contain v_i, since those vectors generate M. Put $L_i = M/M_i$. Under the product quotient map, the submodule generated by v_i has zero coordinates except the ith, where its image is the nonzero submodule L_i. Thus the image contains every direct summand and is the whole direct sum. □

Lemma 5. *Let $P \leq S_m$ be transitive, where $m \geq 4$, acting by coordinate permutation on $V = \mathbb{F}_2^m$. For every invariant subspace $W \leq V$, every P-independent family in W has size at most $m - 2$. For $P = S_m$ the stronger bound two holds.*

Proof. Let M be the submodule generated by an independent family of size s. If $M < V$ and $s \geq m-1$, Lemma 4 forces $\dim M = s = m - 1$ and every simple quotient there has dimension one. Over $\mathbb{F}_2$ a one-dimensional representation is trivial, so the surjection is an isomorphism and M is fixed pointwise by P. But the fixed space of a transitive permutation group on V is the one-dimensional constant subspace, a contradiction.

If $M = V$, at most one of those simple quotients has dimension one: otherwise the product surjection would give two linearly independent maps from V to the trivial module, whereas a P-invariant linear functional has equal coefficients in all m positions. Hence $m \geq 2s-1$, and $s \leq \lfloor (m+1)/2 \rfloor \leq m-2$.

For the final assertion, if every vector in the family is constant, one suffices. Otherwise choose a nonconstant vector v. Subtracting its image under a transposition that exchanges a zero and a one gives $e_i + e_j$. Conjugation then generates the entire even-weight subspace. This subspace has codimension one, so at most one further member of the family is needed. Independence therefore permits at most two. □

Theorem 6. *Every transitive subgroup of $S_2 \wr S_m$ in degree $2m$, $m \geq 5$, has maximum irredundant generating length at most $2m - 4$.*

Proof. Let $W \leq \mathbb{F}_2^m$ be the block kernel and $P \leq S_m$ the transitive block image. The conjugation action on W is coordinate permutation. If $\mu(P) \leq m - 2$, Lemmas 1 and 5 give $\mu(G) \leq 2m - 4$. Otherwise Whiston's equality clause forces $P = S_m$. The sharper module bound gives $\mu(G) \leq (m - 1) + 2 = m + 1 \leq 2m - 4$. □

Lemma 7. *For $m \geq 4$, every transitive subgroup of $S_3 \wr S_m$ has $\mu(G) \leq 3m - 4$. For $m \geq 3$, every transitive subgroup of $S_4 \wr S_m$ has $\mu(G) \leq 4m - 4$.*

Proof. For blocks of size three, write K for the block kernel, $A = K \cap A_3^m$, $W = K/A \leq \mathbb{F}_2^m$, and $P = G/K$. If $P \neq S_m$, then $\mu(P) \leq m - 2$, the binary-module cost is at most $m - 2$, and the ternary-module cost is at most its ambient dimension m. This gives $3m - 4$.

If $P = S_m$, the binary cost is at most two. The ternary cost is at most $m-1$. Indeed, an independent family of m vectors would span all of $\mathbb{F}_3^m$, and Lemma 4 would identify this monomial module with a sum of one-dimensional modules. Its action would then have abelian image. But its permutation image on the coordinate axes is S_m, which is nonabelian. Thus the total cost is at most $(m-1)+2+(m-1) = 2m \leq 3m - 4$. For three blocks of size three use Theorem 3 instead.

Now let the blocks have size four. The block-kernel projections are conjugate. If they are proper subgroups of S_4, every independent family in each projection has at most two members, by Whiston's equality clause. Successive projections of the kernel and Lemma 1 bound its independent families by $2m$. Consequently $\mu(G) \leq (m - 1) + 2m \leq 4m - 4$.

Otherwise every projection is S_4. Put $A = K \cap V_4^m$, where V_4 is the normal Klein four subgroup of S_4. Then G/A has a faithful transitive imprimitive action with m blocks of size three, so the preceding bound gives $\mu(G/A) \leq 3m - 4$. We claim that the normal-module cost of $A \leq \mathbb{F}_2^{2m}$ is at most m. The group $L = K/A$ is a subdirect subgroup of S_3^m. Its normal subgroup $Q = L \cap C_3^m$ projects onto C_3 in every coordinate: the commutator subgroup of L already has this property. Thus Q fixes no nonzero vector in the full $2m$-dimensional module, since the natural C_3 action on each two-dimensional coordinate has no fixed vector. Averaging over the odd-order group Q shows that taking Q-fixed vectors

is exact, so every subquotient of this module also has zero Q-fixed space. In particular every simple G-module quotient in Lemma 4 has dimension at least two. Hence a normal-independent family in A has size at most $\dim A/2 \le m$. The normal-module inequality now gives $\mu(G) \le (3m-4)+m = 4m-4$. □

4 A bounded primitive certificate

For an integer $a > 0$, let $\Omega(a)$ be its number of prime factors with multiplicity. Every strictly ascending subgroup chain in a finite group of order a has at most $\Omega(a)$ strict inclusions, because the product of its successive integral indices divides a. Thus the C-rank is at most $\Omega(|G|)$.

The required program `primitive_bound.g` enumerates the complete GAP primitive-group catalogue in degrees 9–25, omitting A_n, S_n, and reports exactly twelve groups for which this elementary bound exceeds $n-4$. All remaining catalogue entries are thereby discharged. The twelve exceptions are:

$$\begin{array}{r|l} 9 & 3^2:QD_{16}, \quad 3^2:SL_2(3), \quad AGL_2(3), \quad PSL_2(8), \quad PSL_2(8):3; \\ 10 & A_6:2, \quad S_6, \quad A_6 2, \quad A_6:2^2; \\ 11 & M_{11}; \\ 12 & M_{12}; \\ 16 & AGL_4(2). \end{array} \tag{2}$$

The program verifies the precise subgroup and quotient identifications used next. Its retained inputs are the completeness of this catalogue and GAP's exact finite-group algorithms; it performs no search for C-group markings and imports none of the source's reported C-ranks.

For the first two degree-nine groups the irreducible translation module costs one and the quotient has at most four prime factors, giving five. For $AGL_2(3)$ the quotient $GL_2(3)$ has Frattini subgroup of order two and Frattini quotient S_4. Passing to a Frattini quotient preserves irredundant generating families: a subset generates the whole group exactly when its image generates the quotient. Thus $\mu(GL_2(3)) = \mu(S_4) \le 3$, and the affine group has $\mu \le 4$. Every maximal subgroup of $PSL_2(8)$ has order 56, 18 or 14; these have at most four prime factors, so a C-group marking has rank at most five. The extension by 3 is not generated by its involutions: the program checks that they generate only the index-three subgroup.

In degree ten the three groups other than S_6 have normal socle A_6 and quotient of order two or four. Lemma 1 and $\mu'(A_6) \le 4$ give at most six generators; S_6 has at most five. Corollary 2 treats degree sixteen.

For M_{11} the maximal-subgroup orders are $720, 660, 144, 120, 48$. The subgroup generated by all involutions in each has at most six prime factors; in the order-720 case its order is 360. Any C-group parabolic lying in such a maximal subgroup lies in its involution-generated subgroup. Hence M_{11} has C-rank at most seven. For M_{12} the maximal-subgroup orders are

$$7920, 7920, 1440, 1440, 660, 432, 432, 240, 192, 192, 72.$$

The first two are M_{11}. The order-1440 groups have normal A_6 with quotient order four, so their irredundant lengths are at most six. Every remaining maximal subgroup has at most seven prime factors. Deleting one distinguished generator and placing the resulting proper parabolic in a maximal subgroup gives C-rank at most eight for M_{12}. All these finite assertions are checked by the complete program.

For arbitrary degree $n \ge 26$, retain Maróti's Theorem 1.1 in its original form, including the lower containment $A_m^\ell \le G$ in the product-action alternative. If $\ell \ge 2$, the same abstract group acts faithfully on $m\ell$ points, so Whiston bounds its independent families by $m\ell - 1 \le m^\ell - 4 \le n-4$; the inequality holds for $m \ge 2$, $\ell \ge 2$, $m^\ell \ge 9$. If $\ell = 1$, the natural action case is excluded by $G \ne A_n, S_n$; otherwise the k-subset action has $k \ge 2$ and $m \le n/2$, so the same conclusion follows from $m - 1 \le n - 4$. The Mathieu alternative has degree at most 24.

In the remaining alternative, Maróti bounds the group order by $M(n) = n\prod_{i=0}^{\lfloor \log_2 n\rfloor -1}(n-2^i)$. For $26 \le n \le 63$, exact integer arithmetic in the certificate verifies $M(n) < 2^{n-3}$. For $n \ge 64$, put $k = \lfloor \log_2 n \rfloor \ge 6$. Then

$$\log_2 M(n) < (k+1)\log_2 n < (k+1)^2 \le 2^k - 3 \le n-3.$$

The middle inequality follows at $k = 6$ and is preserved when k increases by one. Since C-rank is an integer and is less than $n-3$, it is at most $n-4$. We have proved the full primitive-parabolic bound for all degrees, conditional only on the explicitly retained Whiston and Maróti theorems and the bounded exact catalogue inputs.

5 A uniform simple-kernel bound

Lemma 8. *Let S be a finite nonabelian simple group. Every subdirect subgroup of S^m is a product of diagonal copies of S on a partition of the m coordinates, with automorphisms allowed in the diagonal maps. A strictly increasing chain of subdirect subgroups has at most $m-1$ strict steps.*

Proof. Proceed by induction on m. Project onto the first $m-1$ coordinates. The kernel in the last factor is normal in S, so is either S or trivial. In the first case the subgroup is the product of its projection with S. In the second it is the graph of an onto homomorphism from that projection to S. By induction the projection is a direct product of simple diagonal factors. Their images are commuting normal subgroups of S, so exactly one has nontrivial image, and its map is an isomorphism. The last coordinate therefore joins that diagonal block. This proves the description. In an inclusion of such subgroups the larger group can only refine the partition. If the number of blocks does not increase, their orders are equal and the groups coincide. There are between one and m blocks, proving the chain bound. □

This is the usual subdirect-product lemma, often called Scott's lemma; its complete elementary proof is included here, rather than retained as an unpriced extra premise.

Lemma 9. *Suppose $N \trianglelefteq G$ is a subdirect subgroup of S^m for a finite nonabelian simple group S, and conjugation by G permutes the coordinates transitively, using automorphisms on individual factors. Then*

$$\mu(G) \le \mu(G/N) + \mu'(S) + m - 1. \tag{3}$$

Proof. Normalize an irredundant generating family as in Lemma 1, obtaining $T \cup E$, with T minimally surjecting onto G/N and $E \subset N$. The normalized extra elements stay independent relative to $\langle T\rangle$, by the substitution argument, and no member of T can be dropped since its quotient image is indispensable. Thus this replacement preserves irredundance. Write $T_0 = \langle T\rangle$ and $N_0 = T_0 \cap N$. The subgroup generated by all T_0-conjugates of E is normalized by T_0 and contains E, hence is normal in G. It follows that N is generated by N_0 and these conjugates.

Choose a minimal subset $B \subseteq E$ such that the projections of N_0 and of all T_0-conjugates of B generate S in the first coordinate. Then $|B| \le \mu'(S)$. To see this, for each $b \in B$ choose an element in the projected subgroup generated by its conjugates which does not belong to the subgroup generated by N_0 and the corresponding subgroups for the other members of B. These selected elements form an independent family in S.

Put $J = \langle T, B\rangle$. The group $J \cap N$ projects onto the first copy of S. Conjugation by $N \le S^m$ fixes each coordinate, so the transitive coordinate action of G is already that of T_0. Therefore it projects onto every copy. Add the elements of $E \setminus B$ one by one. Each addition strictly enlarges the generated group, by irredundance. Since the image modulo N stays fixed, it strictly enlarges its intersection with N. These intersections form a chain of subdirect subgroups, with at most $m-1$ strict steps by Lemma 8. Thus $|T| + |E| \le \mu(G/N) + \mu'(S) + m - 1$, as required. □

6 Primitive block stabilizers

Lemma 10. *Let L be primitive of degree $k \geq 5$, with $L \neq A_k, S_k$. Then $\mu'(L) \leq k-3$, except for the degree-six copy of S_5, which has $\mu' = 4$.*

Proof. For $k \geq 26$, the Maróti and Whiston argument above bounds all independent families, not only C-group markings, by $k-4$. For $5 \leq k \leq 25$, the complete primitive catalogue and the prime-factor bound leave exactly the following eleven entries:

$$\begin{array}{r|l} 5 & C_5 : C_4 \\ 6 & A_5,\ S_5 \\ 7 & PSL_3(2) \\ 8 & AGL_3(2),\ PGL_2(7) \\ 9 & AGL_2(3),\ PSL_2(8) : 3 \\ 10 & A_6 : 2^2 \\ 12 & M_{12} \\ 16 & AGL_4(2). \end{array} \tag{4}$$

The same required program checks this complete list and the exact maximal-subgroup, normal-subgroup and quotient facts below.

The group $C_5 : C_4$ has irredundant length at most two by its minimal normal C_5 and cyclic prime-power quotient. Every proper subgroup is cyclic or dihedral of order ten, hence also has bound two. Whiston treats A_5 and S_5 in their natural degrees five. Every maximal subgroup of $PSL_3(2)$ is S_4 or has at most two prime factors, so a generating family has length at most four and a proper subgroup has independent length at most three. The index-two extension $PGL_2(7)$ consequently has independent length at most five. Here and below we use the immediate subgroup version of Lemma 1: $\mu'(G) \leq \mu'(G/N) + \mu'(N)$.

For $AGL_3(2)$ consider any subgroup H, its intersection U with the translation module V, and its linear image $P \leq GL_3(2)$. If $\dim U = 0$, the bound is four. If $\dim U = 1$ or two, P lies in a line or plane stabilizer, each isomorphic to S_4 (its normal 2^2 and quotient $GL_2(2)$ act as the full affine group on four points), so the bound is $3 + \dim U \leq 5$. If $U = V$ and P is irreducible, the normal module costs one and the quotient costs at most four. If P is nontrivial and reducible, it lies in an S_4 stabilizer, and the module costs at most two: three independent vectors would, by Lemma 4, identify V with a sum of three one-dimensional, hence trivial, modules, contradicting the faithful nontrivial action. Finally $P = 1$ costs only three. Thus every H has irredundant length at most five.

Every proper subgroup of $GL_2(3)$ lies in a maximal subgroup whose order has at most four prime factors; the whole group has irredundant length three by its Frattini quotient S_4. Thus $\mu'(GL_2(3)) \leq 4$, and $\mu'(AGL_2(3)) \leq 2 + 4 = 6$. The earlier maximal-subgroup bound for $PSL_2(8)$ gives independent length at most five, so its index-three extension has bound six. The degree-ten group has normal A_6 and quotient of order four, giving at most $4 + 2 = 6$. Every maximal subgroup of M_{12} has at most eight prime factors, so its independent length is at most nine. Finally $AGL_4(2)$ has a four-dimensional normal translation module and quotient A_8, giving $\mu' \leq 4 + 6 = 10$. All satisfy the asserted bounds. □

7 The full imprimitive bound

Theorem 11. *Every transitive imprimitive permutation group of degree $n \geq 9$ has maximum irredundant generating length at most $n-4$.*

Proof. Choose a minimal nontrivial block, of size k, and let $m \geq 2$ be the number of its translates, so $n = km$. The induced group L of the setwise block stabilizer is primitive: a proper block for L would propagate to a smaller block system for G. After choosing identifications of the blocks by elements of

G, the group embeds in $L \wr S_m$. Its block image P is transitive, and the projection K_1 of the block kernel K is normal in L; all other projections are conjugate to it.

Blocks of size two, three or four are covered by Theorems 3, 6 and Lemma 7. Assume $k \geq 5$. If $m \geq 3$ and $L < S_k$, then every subgroup of L has independent length at most $k-2$, by Whiston's equality clause. Successive coordinate projections give $\mu'(K) \leq m(k-2)$, and therefore

$$\mu(G) \leq (m-1) + m(k-2) = n - m - 1 \leq n - 4.$$

If $m = 2$ and $L \neq A_k, S_k$ is not the degree-six copy of S_5, Lemma 10 similarly gives $\mu(G) \leq 1 + 2(k-3) = n - 5$.

It remains to treat $L = A_k, S_k$, and the exceptional degree-six copy of S_5 with $m = 2$. First take $L = A_k$ or S_k. By simplicity of A_k and its trivial centralizer in S_k, the normal subgroup K_1 is either trivial, A_k, or S_k. In the trivial case $K = 1$ and $\mu(G) \leq m - 1 \leq n - 4$. Otherwise $N = K \cap A_k^m$ is subdirect in A_k^m: the commutator subgroup of K already projects onto A_k in every coordinate. The quotient G/N embeds in $C_2 \wr S_m$. Its maximum irredundant length is at most $(m-1) + m = 2m - 1$ by the normal-module bound, so Lemma 9 and $\mu'(A_k) \leq k - 2$ give

$$\mu(G) \leq (2m-1) + (k-2) + (m-1) = k + 3m - 4.$$

For $k \geq 5, m \geq 3$ this is at most $km - 4$, since $(k-3)(m-1) \geq 3$. When $m = 2$, the quotient lies in $C_2 \wr S_2 \cong D_8$; every subgroup of D_8 has irredundant length at most two. The sharper bound is then $\mu(G) \leq 2 + (k-2) + 1 = k + 1 \leq 2k - 4$.

Finally let $k = 6, m = 2, L \cong S_5$. Use the characteristic subgroup $A_5 \lhd L$ and the same argument in the abstract factors. The block kernel is trivial or projects onto A_5 or S_5. In the latter two cases Lemma 9, now with $S = A_5$, gives $\mu(G) \leq 2 + 3 + 1 = 6 \leq 12 - 4$. This completes all cases and every parameter value. □

Corollary 12. *Let $(S_n; \rho_0, \dots, \rho_{r-1})$ be a C-group with $n \geq 9$ and $r \in \{n-2, n-1\}$. Every maximal standard parabolic is intransitive on the n letters.*

Proof. Each such parabolic has an independent distinguished family of $r - 1 \geq n - 3$ elements. The primitive bound and Theorem 11 exclude every transitive proper subgroup except A_n. Suppose one parabolic is A_n and let a be the omitted involution. For each other distinguished involution b, the intersection property says $A_n \cap \langle a, b \rangle = \langle b \rangle$. Since $aba \in A_n$, this forces $aba = b$. Thus a centralizes all of A_n and $S_n = A_n \times \langle a \rangle$, impossible. The centralizer of A_n in S_n is trivial for $n \geq 4$, as is seen by commuting with all three-cycles. Hence no maximal parabolic is transitive. □

8 A finite recognition interface

For a partition π of a finite set Q, write $\Upsilon(\pi) = \prod_{B \in \pi} \mathrm{Sym}(B)$, as a subgroup of $\mathrm{Sym}(Q)$. Let $(h_i)_{i \in I}$ be a finite family in $\mathrm{Sym}(Q)$ and let $H_S = \langle h_i : i \in S \rangle$ for $S \subseteq I$. Fix $d \in Q$. Suppose each S is assigned a partition π_S of Q. If $x \notin Q$, let π_S^x be the partition of $Q \cup \{x\}$ obtained by adjoining x to the block containing d.

We impose the following finite conditions, for every $S, R \subseteq I$:

$$H_S \cap H_R = H_{S \cap R}, \qquad H_S \cap \Upsilon(\pi_R) = H_{S \cap R}, \tag{5}$$

$$\pi_S \wedge \pi_R = \pi_{S \cap R}, \qquad \langle H_S, (d\ x) \rangle = \Upsilon(\pi_S^x). \tag{6}$$

Here the meet is common refinement. In particular, taking $S = R$ in (5) gives $H_S \leq \Upsilon(\pi_S)$. These hypotheses concern subgroups on $|Q|$ or $|Q| + 1$ letters; they do not assume the desired intersection property on any larger degree.

Let T be a finite tree meeting Q in exactly $\{d\}$, with at least one edge, and suppose d is a leaf of T. Put $\Omega = Q \cup V(T)$. Alongside the h_i, take the transposition t_e exchanging the endpoints of each edge e of T. All these permutations fix the unspecified letters.

Theorem 13. *Under* (5)–(6)*, the family* $(h_i)_{i\in I}, (t_e)_{e\in E(T)}$ *has the full subset intersection property. If the* h_i *are distinct nonidentity involutions, then so is this enlarged family and it is a C-group marking of its generated group. If moreover* $\pi_I = \{Q\}$*, that group is* $\mathrm{Sym}(\Omega)$*.*

Proof. For $F \subseteq E(T)$, let D_F be the vertex set of the component containing d in $(V(T), F)$; the other component sets will be denoted C. The tree transpositions on a connected component generate its full symmetric group: along the unique paths, conjugating endpoint transpositions produces every transposition from one fixed vertex. Thus

$$\langle h_i\ (i \in S), t_e\ (e \in F)\rangle = H_S[D_F] \times \prod_{C\neq D_F} \mathrm{Sym}(C), \tag{7}$$

where $H_S[\{d\}] = H_S$. If $D_F \neq \{d\}$, then $H_S[D_F]$ is the Young subgroup obtained from π_S by adjoining $D_F \setminus \{d\}$ to its d-block. Indeed, choose any $x \in D_F \setminus \{d\}$. The subgroup $\mathrm{Sym}(D_F)$ contains $(d\ x)$, so (6) supplies every transposition within each unchanged core block and within the enlarged d-block on $Q \cup \{x\}$. Conjugation by $\mathrm{Sym}(D_F)$ supplies the same transpositions for every point of D_F. Conversely all the original generators preserve the claimed blocks. This proves (7).

We now intersect the groups for (S, F) and (R, E). Intersections of connected subtrees of a tree are connected or empty. More precisely, two vertices belong to one component in both forests if and only if the unique path between them belongs to $F \cap E$. Hence common refinement of their component partitions is the component partition of $F \cap E$. This proves the classical forest-intersection argument used throughout the present construction; it is also the argument of our earlier UB.F.

If $D_F = D_E = \{d\}$, the core intersection is $H_{S\cap R}$ by (5); all other factors follow from the forest argument. Suppose exactly one of D_F, D_E is a singleton, say $D_F = \{d\}$. Every permutation in the first group preserves Q setwise and acts there in H_S. Its additional membership in the second group restricts this action exactly to $\Upsilon(\pi_R)$. Equation (5) gives $H_{S\cap R}$ on Q. Off Q, the two forest component partitions have the refinement just described; their intersections with the enlarged root component are connected subtrees too. Consequently the remaining factors are exactly the symmetric groups on the nonroot components of $F \cap E$.

Finally suppose both root components are nonsingletons. The unique edge incident with d lies in both F and E. Therefore $D_{F\cap E} = D_F \cap D_E$ is also nonsingleton. Both groups in (7) are now Young subgroups of $\mathrm{Sym}(\Omega)$. Their intersection is the Young subgroup of the common refinement. On the core this is $\pi_S \wedge \pi_R = \pi_{S\cap R}$; the root block acquires exactly $D_F \cap D_E \setminus \{d\}$, and all other blocks are the nonroot components of $F \cap E$. This is precisely (7) for $(S \cap R, F \cap E)$. All subset intersections have been proved.

No tree transposition is supported on Q, and different edges give different transpositions. Thus the distinctness assertion follows. For the whole edge set, $D_F = V(T)$ is nonsingleton; if π_I has one block, (7) is $\mathrm{Sym}(\Omega)$. □

9 A uniform presentation for a tree

Let T be a finite tree on n vertices. Write s_e for its edge transpositions. The following is the presentation in Fernandes–Leemans, Proposition 3.1, without its unnecessary restriction $n \geq 7$.

Theorem 14. *The symmetric group on the vertices of T has generators* s_e *and relations*

$$s_e^2 = 1, \qquad (s_e s_f)^2 = 1 \quad (e, f \textit{ disjoint}), \qquad (s_e s_f)^3 = 1 \quad (e, f \textit{ meet}),$$

together with $(s_e s_f s_e s_g)^2 = 1$ *for every three distinct edges* e, f, g *incident with one vertex. For each unordered triple it is enough to impose the relation in one ordering; all the other orderings then follow.*

Proof. First observe that the choice of ordering is immaterial. For three incident involutions a, f, g, the pairwise braid relations give

$$[afa, g] = 1 \iff [a, fgf] = 1 \iff [a, gfg] = 1 \iff [aga, f] = 1.$$

The first and last equivalences follow by conjugation and a braid identity. Also $afa = faf$ interchanges the first two letters. These interchanges generate all orderings of the triple.

Let $P(T)$ be the presented group. The indicated transpositions satisfy the relations and generate $\mathrm{Sym}(V(T))$, so there is a surjection $P(T) \to \mathrm{Sym}(V(T))$. We prove $|P(T)| \le n!$ by induction on n. The cases $n \le 2$ are immediate. Remove a leaf p, with neighbor q, and write $a = s_{pq}$. By induction the subgroup H generated by the remaining edge generators is a quotient of $\mathrm{Sym}(V(T)\setminus\{p\})$. Its displayed permutation image is that entire symmetric group, so the quotient is faithful and we may identify H with S_{n-1}.

We first prove that a centralizes H_q, the point stabilizer of q in H. It commutes with every edge not incident with q. If $f = s_{qu}$ and $g = s_{qv}$ are two distinct other edges at q, the triangle relation says that afa commutes with g. The braid identity $afa = faf$ turns this into the assertion that a commutes with $fgf = (uv)$. The edge transpositions within the components of $T - \{p, q\}$, together with these transpositions between their neighbors at q, form a connected transposition graph on all vertices other than p, q. They therefore generate H_q. This proves the assertion, including the case where there is at most one such component.

Choose a neighbor u of q in $T - p$. The relation for a and s_{qu} is the braid relation. Conjugating it by H_q, which acts transitively on the remaining $n - 2$ vertices and centralizes a, gives

$$af_v a = f_v a f_v \qquad \text{for } f_v = (qv), \quad v \ne p, q.$$

Set $f_q = 1$. Every element of H has a unique left coset of H_q, represented by some f_v. Consequently the union of the following at most n left cosets of H is stable under right multiplication by H:

$$H, \qquad Haf_v \quad (v \ne p).$$

Explicitly, $f_v h = k f_w$ for some $k \in H_q$, so $Haf_v h = Hakf_w = Haf_w$. It is stable under multiplication by a as well: $Ha \cdot a = H$, and for $v \ne q$ the braid identity gives $Haf_v a = Hf_v af_v = Haf_v$. The union contains the identity, and $P(T)$ is generated by H and a. It is therefore all of $P(T)$, whence $[P(T) : H] \le n$ and $|P(T)| \le n!$. The surjection to S_n is an isomorphism. □

10 Transporting an exact presentation

Lemma 15. *Suppose $K = \langle s_e \mid R(s)\rangle$. Let $r_i = v_i(s)$ generate K and choose words $w_e(r)$ with $w_e(v(s)) = s_e$ in K. Then an exact presentation on the marking r_i is*

$$\langle r_i \mid R(w(r)), \quad r_i = v_i(w(r)) \text{ for every } i\rangle. \tag{8}$$

Proof. The prescribed elements of K satisfy all displayed relations, giving a surjective map from (8) to K. Conversely $s_e \mapsto w_e(r)$ defines a homomorphism from K to (8), because every relator $R(w)$ is imposed. One composite fixes each s_e by the word identity in K; the other fixes each r_i by the second group of relations. Thus these homomorphisms are inverse. □

11 Attribution, retained inputs and formal scope

The normal-quotient inequality is Cameron–Cara's Theorem 1.1, not a new result. Its full proof and the elementary subdirect-product lemma are included. Forest transpositions, tree presentations and changes of generators are classical mechanisms; the source application and its historical attribution are identified in the companion paper. The module bounds and the finite-core interface are the shared arguments used here to reorganize that recovery. We make no historical priority claim for them.

Two substantial classification inputs remain external. Whiston's independent-family bound for symmetric groups, including its equality clause, is retained from [W], as stated and used in the genuine

authored paper [CC, §§1–2]. The original [W] full text has not been obtained for this packet. The companion also retains the three possibilities for a one-generator omission from a maximum family, stated in the proof of [CC, Theorem 2.1]. Maróti's original Theorem 1.1 [M] is retained with both containments in its product-action case. These results depend on finite-group classification; their proofs are not replaced by the elementary arguments here.

The entire `primitive_bound.g` is a required finite premise: it uses the complete GAP primitive catalogue through degree 25 and exact maximal-subgroup, socle and quotient algorithms. Its exception lists, all special identifications and integer inequalities are explicit above. Completeness of that catalogue and correctness of those group algorithms are retained computational inputs. This is not an exhaustive search over C-group markings. The companion's core and local-obstruction programs are finite validation of fully written arguments, with their roles kept distinct in the complete program appendix.

The eight declarations in `IrredundantNormalization.lean` prove the actual normalization leg of Lemma 1: quotient-kernel membership, preservation of generated subgroup and independence of the normalized extra elements, first under base-group conjugation and then under all conjugations for an abelian normal kernel. The latter theorem explicitly assumes $G = N\langle T\rangle$ and internal commutativity of N. It does not assume the desired independence conclusion. The Lean file does not prove the cardinal bounds, irreducibility consequences, Scott's lemma, the primitive classification, finite catalogue completeness, the attachment theorem or any source-specific geometry. These remain the written and declared computational obligations. Fresh compilation reports standard Lean axioms only; cached pinned library objects are part of the trusted formal environment.

References

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[W] J. Whiston, Maximal independent generating sets of the symmetric group, *J. Algebra* **232** (2000), 255–268, doi:10.1006/jabr.2000.8399.

[GAP] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version used in the saved replay. Primitive-group catalogue and exact finite-group routines.